ALI WAHAJ

Software Engineer — Backend & Infrastructure — Python — Cloud & DevOps — AI
(+92)3314459672|https://www.linkedin.com/in/syed-wahaj|wahajali98@gmail.com|https://github.com/WahajAli14

EDUCATION & CERTIFICATIONS

B.Sc. Computer Science Major **2020 – 2024**
Concentration Software Engineering
FAST NUCES University CGPA:3.35/4

DeepLearning.AI Machine Learning Specialization **Dec 2024**
Microsoft Azure Fundamentals (AZ-900) **Dec 2023**

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++, YAML, .NET
Frontend: React.js, Angular, Tailwind
Backend: AWS Lambda, Serverless Framework, .NET, FastAPI, SQLAlchemy
Cloud & DevOps: AWS, Terraform, Docker, Kubernetes, GitHub Actions, AWS SQS
Databases & Caching: SQL Server, MySQL, MongoDB, Redis
AI / LLM Applications: RAG, LangChain, ChromaDB, Vector Databases

EXPERIENCE

Backend Software Engineer **April 2025 - Present**
LinkedMatrix (Onsite)
Worked on a serverless U.S-based laundry care platform supporting customer orders, pickup and delivery workflows, worker operations, pricing, client accounts, address management, and masked customer-worker communication.

- Optimized high-volume backend endpoints by improving SQLAlchemy queries and removing **N+1 issues**, reducing response time by **70% from 20s to 6s**
- Reduced Mapbox API costs by storing reusable distance records, cutting processing time by **30%**, reducing **100K+ monthly API calls** to near zero, and expiring records after **3 months** to keep location data updated.
- Reduced proxy messaging costs by **50%** through phone-number validation and logic preventing repeated messages to non-responsive numbers.
- Implemented Redis caching for frequently accessed backend data, improving selected workflow performance by approximately **50%**.
- Implemented **AWS SQS-based async processing** for heavy reports and worker image-change workflows, reducing API blocking and improving retry reliability.
- Designed **normalized** database structures to support workflow changes, improving query efficiency, maintainability, and long-term scalability.
- Implemented **Twilio Proxy phone masking** for secure customer-worker SMS/call communication in a serverless platform using **Python, AWS Lambda, Serverless Framework, SQLAlchemy**
- Planned and executed AWS RDS MySQL upgrade from version 8.0 to 8.4, including compatibility assessment, snapshot creation, parameter-group updates, upgrade execution, post-upgrade validation, and rollback preparation.
- Upgraded Keycloak from version 20 to version 25, resolving compatibility issues, updating configurations, validating authentication workflows, and ensuring stable integration with existing backend services.
- Migrated the platform's tax calculation integration from TaxJar to TaxCloud, reducing tax-processing costs by approximately **30%** while maintaining accurate tax calculation and compliance workflows.

Associate Software Engineer **July 2024 - March 2025**
CureMD (Onsite)
Worked on healthcare EDI transaction modules related to eligibility, ERA, and claims processing, supporting enterprise healthcare workflows:

- Optimized stored procedures by removing duplicate data handling and improving query logic, reducing response time from **8s to under 1s**.
- Optimized backend endpoints and database queries for the **11x project**, reducing large-volume data retrieval time from **17s to under 3.5s**.
- Collaborated with cross-functional teams to analyze multiple query paths, minimize duplicate data retrieval, and deliver optimized results in significantly reduced processing time.
- Worked on healthcare EDI modules for **eligibility, ERA, and claims processing**, supporting enterprise healthcare transaction workflows.
- Extracted and Processed **ERA/EDI** data, improving backend accuracy and transaction reliability.
- Built reusable Angular components and optimized TypeScript business logic to improve frontend modularity and lag.
- Developed data verification utilities and automated workflows, reducing manual validation effort by ~8 hours/week

Machine Learning Internship **May - June 2023**
TechnoColab (Remote)

- Performed EDA, data cleaning, feature engineering, visualization, and dataset preparation in Python, while testing and comparing multiple machine learning models to identify the best-performing solution.